

General relativity BIGHW-3

PROBLEM 1. Consider an inflation scenario based on GR with a canonical twoderivative scalar field. Corresponding equations read:

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0 \text{ and } 3H^2 = \frac{1}{2}\dot{\phi}^2 + V(\phi)$$

Here $H = \dot{a}/a$ is the Hubble parameter, $a = a(t)$ is the scale factor and dot is a derivative with respect to time. Show that if the so called first and second slow-roll parameters are small, that is

$$\varepsilon = -\frac{\dot{H}}{H^2} \ll 1 \text{ and } \eta = -\frac{\ddot{\phi}}{H\dot{\phi}} \ll 1$$

then inflation can happen and lasts sufficiently long. Namely, that the scale factor is approximately de Sitter one, i.e. $a \sim e^{Ht}$ and the number of e -folds can reach $N = 60$. Here N is introduced as $dN = Hdt$.

SOLUTION:

The energy momentum tensor of scalar field ϕ

$$\rho = \frac{1}{2}\dot{\phi}^2 + V(\phi)$$

$$p = \frac{1}{2}\dot{\phi}^2 - V(\phi)$$

Then by Friedmann equations

$$H^2 = \frac{1}{3}(\frac{1}{2}\dot{\phi}^2 + V), \frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p) = H^2(1 - \frac{1}{2}\frac{\dot{\phi}^2}{H^2})$$

and the conservation of energy-momentum tensor gives

$$\frac{d\rho}{dt} + 3H(p + \rho) = 0 \implies \ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0$$

When the slow-roll conditions $\dot{\phi}^2 \ll V(\phi)$, $|\ddot{\phi}| \ll |3H\dot{\phi}|$, $|\frac{dV}{d\phi}|$ (which are equivalent to two slow-roll parameters are small), then

$$H^2 \sim \frac{1}{3}V(\phi), \dot{\phi} \sim -\frac{1}{3H} \frac{dV}{d\phi}$$

and

$$\dot{H} = -\epsilon H^2 \implies \frac{1}{H} - \frac{1}{H_0} = \epsilon t \implies H = \frac{H_0}{\epsilon H_0 t + 1}$$

Then

$$a(t) \sim \exp\left(\int H dt\right) \sim \exp\left(\int \frac{H_0}{\epsilon H_0 t + 1} dt\right) \sim (1 + \epsilon H_0 t)^{\frac{1}{\epsilon}}$$

When ϵ is small

$$(1 + \epsilon H_0 t)^{\frac{1}{\epsilon}} \sim e^{H_0 t}$$

In order to solve horizon problem and let microwaves before inflation causally connected with observation, we have to let

$$\frac{a_1 H_1}{a_3 H_2} = e^N \frac{a_1 H_1}{a_2 H_2} > 1$$

where 3 represents the start of inflation, 2 represents the end of inflation and 1 represents now. Suppose our universe after the inflation is based on radiation, then $a \sim H^{-\frac{1}{2}}$, $e^N \frac{a_1 H_1}{a_2 H_2} = e^N \left(\frac{H_1}{H_2}\right)^{\frac{1}{2}} > 1$ where $H_1 \sim 10^{-33} eV$ is the Hubble parameter now and $H_2 \sim 10^{15} GeV = 10^{24} eV$. Then

$$N > \frac{1}{2} \ln \frac{H_2}{H_1} \sim 66$$